

SLANGKOP, South Africa

WMO: 689120

Lat: 34.1483S

Lon: 18.3192E

Elev: 23

StdP: 14.68

Time Zone: 2.00 (E02)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	48.9	50.1	35.2	30.1	64.1	38.6	34.5	57.9	38.1	56.6	34.2	56.5	4.9	140	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.2	78.8	64.1	76.2	63.7	73.9	62.8	67.3	74.5	66.0	72.6	65.0	71.1	15.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	91.7	70.6	63.6	88.0	69.3	62.6	84.9	68.1	31.8	74.7	30.8	72.9	30.0	71.1	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.2	28.9	25.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.1	65.8	66.0	64.1	61.9	59.7	58.0	57.1	56.6	57.7	60.4	62.0	64.7
	DBStd	4.77	3.46	3.68	3.56	4.13	3.25	2.97	3.71	2.73	3.11	3.56	3.58	3.52
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1650	29	26	61	112	170	212	250	262	223	151	106	48
	CDD50	4063	491	449	436	356	300	239	220	204	231	322	359	456
	CDD65	238	56	55	32	18	6	2	4	1	3	8	14	39
	CDH74	510	117	106	64	62	23	5	11	2	13	17	26	64
	CDH80	49	7	11	8	10	1	0	0	0	3	0	1	8
Wind	WSAvg	11.8	13.1	11.9	11.1	10.0	10.1	12.6	11.7	11.8	11.9	11.7	12.5	12.8
Precipitation	PrecAvg	27.7	0.9	0.6	0.9	2.1	3.7	4.2	4.6	4.5	2.6	1.4	1.3	0.8
	PrecMax	36.2	3.9	1.4	3.0	5.1	6.9	7.3	9.9	8.1	6.3	5.1	3.9	1.8
	PrecMin	18.3	0.0	0.0	0.0	0.5	0.2	1.1	1.7	2.1	0.9	0.0	0.1	0.0
	PrecStd	5.0	0.9	0.4	0.7	1.1	1.5	1.3	2.0	1.9	1.3	1.0	1.1	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.7	80.8	80.3	82.6	77.1	73.5	75.5	72.4	75.1	77.2	77.4	78.9
		MCWB	66.6	67.1	65.1	61.8	59.8	56.8	55.9	53.2	58.4	60.9	61.4	66.3
	2%	DB	77.4	77.0	75.1	74.5	70.2	67.2	70.6	65.9	66.8	71.6	72.8	75.3
		MCWB	66.2	65.7	63.7	61.7	58.9	55.3	53.9	55.5	57.1	60.3	61.2	64.6
	5%	DB	74.2	74.2	72.1	70.1	66.6	64.4	65.9	62.7	63.8	68.6	69.6	72.5
		MCWB	64.3	65.0	63.0	60.9	58.3	56.7	55.4	55.6	56.7	59.4	60.1	63.2
	10%	DB	71.6	71.5	69.7	67.7	64.2	62.2	62.0	60.7	61.7	66.0	67.3	70.1
		MCWB	63.7	63.7	62.3	60.1	58.3	56.4	55.7	55.1	55.9	58.6	59.6	62.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.0	69.4	67.4	64.9	62.7	60.6	59.1	59.4	60.2	63.4	64.4	67.6
		MCDB	77.4	76.7	73.5	76.0	67.1	64.1	64.3	64.4	67.7	71.3	72.4	76.1
	2%	WB	67.2	67.2	65.5	63.5	61.2	59.2	58.1	57.9	58.5	61.7	62.4	66.0
		MCDB	74.4	73.9	71.3	70.1	65.4	62.6	62.8	62.1	63.8	68.6	68.9	73.0
	5%	WB	65.7	65.8	64.4	62.2	60.1	58.2	57.3	56.7	57.6	60.3	61.3	64.6
		MCDB	72.1	71.8	70.4	67.6	64.1	61.6	61.5	60.4	62.1	66.0	67.3	70.5
	10%	WB	64.6	64.8	63.4	61.1	59.2	57.4	56.4	55.8	56.7	59.2	60.3	63.5
		MCDB	70.3	70.3	68.6	66.0	62.8	60.7	60.6	59.4	60.7	64.2	65.8	68.9
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.2	9.1	8.9	7.8	7.3	7.8	7.2	7.4	8.1	8.1	8.9
		MCDBR	13.7	13.7	13.7	13.4	12.7	11.7	14.6	11.2	11.6	12.5	12.5	12.7
		MCWBR	7.0	6.8	6.4	6.4	6.3	6.8	6.7	7.3	6.5	6.4	6.2	6.7
	5% WB	MCDBR	11.7	11.8	10.7	10.5	9.3	8.1	9.4	8.8	9.0	9.7	10.5	11.1
		MCWBR	7.2	6.9	6.4	6.5	6.0	6.0	6.5	6.6	6.1	6.0	6.3	6.7
Clear-Sky Solar Irradiance	taub	0.346	0.348	0.337	0.326	0.325	0.320	0.321	0.335	0.345	0.355	0.342	0.347	
	taud	2.554	2.557	2.582	2.607	2.597	2.584	2.576	2.536	2.470	2.460	2.538	2.539	
	Ebn at Noon	314	307	299	285	267	260	266	277	291	301	313	315	
	Edh at Noon	34	33	30	26	23	22	23	27	33	36	35	35	
All-Sky Solar Radiation	RadAvg	2573	2290	1831	1340	930	781	870	1110	1535	2020	2404	2610	
	RadStd	60	70	75	65	56	46	50	53	62	91	74	60	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+0.65	+0.88	N/A	N/A	+0.95	+0.85	-4	-133	+200	+110	

Nomenclature: See separate page